

COMMENTARY

Estimating effective sample sizes using standard errors from appropriate analyses of cluster-randomized trials

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Abstract

Background and Objectives: Cluster-randomized trials (CRTs) typically provide less information than similarly sized trials with individual randomization due to correlation within clusters, ie, the effective sample size (ESS) is typically lower than the total number of participants. Ignoring clustering in analyses can lead to overly precise results (too small *P* values and too narrow confidence intervals) and bias. ESSs from CRTs can be valuable for interpretation or meta-analysis. ESSs can be estimated using intraclass correlation coefficients (ICCs), but these are often not known and must be assumed.

Methods: Two approaches for estimating ESSs from CRTs using standard errors from appropriate analyses taking clustering into account, but not requiring ICCs, are presented. The first approach scales event and participant counts by the ratio between the variances from appropriate analyses and simple analyses not taking clustering into account. The second approach uses an optimization procedure to modify event proportions or group means and counts. Code and explanations to implement both approaches are provided.

Results: Using data from three illustrative example CRTs, both approaches are compared to not taking clustering into account and results from appropriate analyses. Both approaches avoid inappropriately high precision when group-level summary data are analyzed. The second approach also corrects potential bias in estimates that can be introduced by ignoring clustering.

Conclusion: These approaches may be used to aid interpretation and evidence synthesis of CRTs and can be used where results from appropriate analyses of CRTs are presented, even if ICCs are not reported. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Clinical trials; Cluster-randomized trials; Effective sample sizes; Analysis; Meta-analysis; Clustering

1. Introduction

Randomized trials typically randomize individuals independently of each other. In cluster-randomized trials (CRTs), groups of participants (eg, hospital departments) are randomly assigned to the same intervention parallelly (parallel-arm CRTs) [1]. In extensions of CRTs, the allocation may be changed one or more times at the cluster-level (cluster-crossover trials), or clusters may be randomly allocated to switch from usual practice to a new intervention at different time points (stepped-wedge trials) [1]. CRTs may be used when individual randomization is not meaningful

(eg, for education interventions aimed at clinicians in different departments) or due to logistical benefits or benefits due to simplified consent procedures [1].

Common to CRTs is that the *unit of allocation* (clusters) differs from the *unit of analysis* (individual outcomes) [1,2]. Because participants within clusters typically are more similar and have more similar outcomes than participants in other clusters, outcomes within clusters are correlated and not statistically independent. This can be quantified by the *intraclass correlation coefficient* (ICC) [1–3]. Due to correlation, the information provided by a CRT is generally smaller than that provided by the same number of individually randomized participants [1,2], meaning that more participants are required to obtain the same certainty in a CRT compared to a trial with individual randomization. In other words, the *effective sample size* (ESS; the number of individually randomized participants that would provide the same information as the CRT) is

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What is new?**Key findings**

- Two approaches estimate effective sample sizes using SEs from cluster trials.
- Both adjust participant/event counts; the second adjusts estimates to handle bias.

What this adds to what is known?

- Both can be used when intra-cluster correlation coefficients are not available.

What is the implication and what should change now?

- Consider these approaches when appropriate SEs from cluster trials are reported.

lower than the actual number of participants. To avoid incorrect conclusions due to incorrectly narrow confidence intervals (CIs) and incorrectly small P values, CRTs should be analyzed using appropriate methods (eg, generalized linear mixed-effects models with random effects for clusters or generalized estimating equations) [1,2]. Analyses of CRTs not taking clustering into account can also lead to bias, particularly if cluster sizes vary [4,5].

The inclusion of CRTs in meta-analyses comes with specific challenges [2]. Meta-analyzing group-level summary data (eg, participant numbers and event counts for binary outcomes or means and SDs for normally distributed continuous outcomes) directly can lead to invalid results [2,6], as this ignores clustering. If an analysis appropriately handling clustering has been conducted, the point estimate and standard error (SE) can be meta-analyzed using the generic inverse variance approach [2]. However, this requires that appropriate estimates and SEs from all included CRTs are available using the same summary effect measure (eg, risk ratio [RR], odds ratio [OR], or risk difference [RD]).

Even when appropriate analyses of CRTs are available, it may be valuable to calculate ESSs to aid interpretation and evidence synthesis, including in evidence summaries [7]. Meta-analyses often use group-level summary data, as these are more intuitively interpreted than SEs, and some meta-analytical methods (eg, trial sequential analysis [8]) do not support the generic inverse variance approach. For CRTs, ESSs may be calculated by dividing the actual numbers with the *design effect*, followed by meta-analysis using conventional methods for group-level summary data with the corrected participant/event counts [2,3]. The design effect-approach is widely recommended [2], but requires an estimate of the ICC for each outcome and the mean cluster size ($\text{design effect} \approx 1 + (\text{mean cluster size}$

$- 1) * \text{ICC}$) [2], and calculations may be incorrect with few clusters or substantially varying cluster sizes [3]. Despite recommendations [9], ICCs are often not reported [9–11], and when available, they are often only reported for the primary outcome. When ICCs are not available, they are typically assumed based on ICCs from similar trials [11,12]. However, even for trials in the same setting, ICCs can vary substantially across trials and outcomes [13,14], and incorrectly assumed ICCs may affect ESSs and thus interpretation or meta-analysis. Furthermore, for extensions of parallel-arm CRTs and CRTs with additional layers of clustering, design effect formulae are more complex and require additional correlations [3], which may similarly not be available. Finally, the design effect approach only handles the precision/power issue but does not correct any potential statistical bias.

Herein, two alternative approaches for estimating ESSs are presented. Both approaches require the results from an appropriate analysis taking clustering into account without requiring ICCs, and both approaches may be used for both parallel-arm CRTs and extensions. The approaches are described along with statistical code to apply them to common summary effect measures, and examples of their use on results from actual CRTs.

2. Methods

2.1. General methodological considerations

Both approaches require the following:

- (1) Group-level summary data for each group: event counts and total participant numbers for RRs, ORs, and RDs; and means, SDs, and total participant numbers for mean differences and ratios of means.
- (2) Results of an appropriate analysis taking clustering into account (SE and point estimate).

SEs may be calculated from either CIs or P values [15,16]; SEs of ratio measures (RRs, ORs, and ratios of means) should be on the logarithmic scale. Both approaches use established formulae for calculating SEs for simple, unadjusted analyses of group-level summary data [17,18] (Table 1).

Appendix A contains user-friendly R functions to implement both approaches, estimate SEs from CIs or P values, and to calculate summary measures with CIs and P values from group-level summary data; functions involving CIs/ P values use large sample/normality approximations. Appendix B provides additional explanation and examples using the code. Code/analyses used R v4.4.1.

2.2. Approach #1: using the ratio between variances

Multiplying all total participant and event counts by a constant in the formulae in Table 1 will change the variance

Table 1. Formulae for calculating summary measures and standard errors

Summary measure	Estimate	Standard error ^a
RR	$(E_i/N_i)/(E_c/N_c)$	$\sqrt{1/E_i - 1/N_i + 1/E_c - 1/N_c}$
OR	$(E_i/(N_i - E_i))/(E_c/(N_c - E_c))$	$\sqrt{1/E_i + 1/(N_i - E_i) + 1/E_c + 1/(N_c - E_c)}$
RD	$(E_i/N_i) - (E_c/N_c)$	$\sqrt{(E_i * (N_i - E_i))/N_i^3 + (E_c * (N_c - E_c))/N_c^3}$
MD	$M_i - M_c$	$\sqrt{(SD_i^2/N_i) + (SD_c^2/N_c)}$
RoM	M_i/M_c	$\sqrt{1/N_i * (SD_i/M_i)^2 + 1/N_c * (SD_c/M_c)^2}$

Formulae for calculating summary measure estimates and standard errors according to Schwarzer et al. (RRs, ORs, RDs, and MDs) [17] and Friedrich et al. (RoMs) [18].

c, denotes the control group; E, denotes event counts; i, denotes the intervention group; M, denotes means; MD, mean difference; OR, odds ratio; N, denotes participant numbers; RD, risk difference; RoM, ratio of means; RR, risk ratio (relative risk); sqrt, square root.

^a For relative summary effect measures (RRs, ORs, and ROMs), the standard errors are on the logarithmic scale.

(SE²) by division with the same factor; eg, halving all participant/event counts will double the variance.

ESSs can thus be approximated by these steps:

- (1) Extract the SE from an appropriate analysis taking clustering into account (on the logarithmic scale for ratio measures).
- (2) Calculate the SE using raw group-level summary data without accounting for clustering.
- (3) Calculate the ratio between variances:

$$\text{Conversion factor} = \text{SE}_{\text{simple}}^2 / \text{SE}_{\text{cluster}}^2$$

- (4) Multiply all participant counts and event counts (for binary outcomes) by the conversion factor (for normally distributed outcomes, means and SDs are unchanged).

Approach #1 is simple, and calculations can be done manually or using functions from [Appendix A](#). Approach #1 will correct ESSs to avoid overly precise results, ie, *P* values will not be incorrectly too small and CIs will not be incorrectly too narrow. Potential bias in estimates due to not taking clustering into account will not be handled by approach #1 (similarly to when ESSs are estimated using ICCs). Consequently, if approach #1 is used, it is sensible to check if point estimates from the appropriate analysis taking clustering into account and the analysis using group-level summary data after correction are identical. If not, approach #2 may be preferable.

2.3. Approach #2: optimization procedure

The second approach uses an optimization procedure and the following steps:

- (1) Extract the point estimate and SE from an appropriate analysis taking clustering into account (SE on the logarithmic scale for ratio measures).
- (2) Calculate the proportions of participants in each group using the raw group-level summary data. Alternatively, proportions can be set to fixed values (eg,

50%:50%) matching the intended allocation proportions, even if the actual allocation proportions are different due to different cluster sizes. While calculated/fixed proportions will affect adjusted counts differently, results obtained by analyzing the adjusted data will correspond to those from the original appropriate analysis regardless.

- (3) Adjust the event proportions (binary outcomes) or means (continuous outcomes) in each group by similar amounts using an optimization procedure (by subtracting a specific value from one group and adding it to the other group) until the point estimate obtained from an analysis not taking clustering into account matches the point estimate from the appropriate analysis.
- (4) Estimate the total ESS using an optimization procedure with the calculated/fixed participant proportions in each group (step #2) and the adjusted event proportions or means in each group (step #3) by changing the total ESS until the SE from a simple analysis not taking clustering into account matches the SE from the appropriate analysis.
- (5) Calculate participant counts in each group by dividing the estimated total ESS with the allocation proportions. For binary outcomes, calculate the event counts by multiplying the participant counts in each group with the adjusted event proportions. For normally distributed outcomes, adjusted means in each group are used, with group SDs being unchanged.

Approach #2 is somewhat more advanced than approach #1 but is easily carried out using the functions in [Appendix A](#).

2.4. Illustrative comparison

Both approaches are illustrated and compared to results from appropriate analyses taking clustering into account and analyses of group-level summary data ignoring clustering using outcomes from three CRTs specifically chosen to be illustrative (ie, results from the different approaches vary):

- (1) The secondary outcome *clinically important upper gastrointestinal bleeding* from the Proton Pump

Inhibitors vs Histamine-2 Receptor Blockers for Ulcer Prophylaxis Treatment in the Intensive Care Unit (PEPTIC) trial, a cluster-crossover trial comparing proton pump inhibitors (intervention) versus histamine-2 receptor blockers (control) for stress ulcer prophylaxis in adult intensive care unit (ICU) patients on invasive mechanical ventilation across 50 clusters (ICUs) [19].

- (2) The secondary outcome *days alive and free of mechanical ventilation within 90 days* from the Selective Decontamination of the Digestive Tract in Intensive Care Unit (SuDDICU) trial, a cluster-crossover trial comparing selective decontamination of the digestive tract versus standard care in adult mechanically ventilated ICU patients across 19 clusters (ICUs) [20].
- (3) One of the two co-primary outcomes, *unpredicted difficult mask ventilation*, from the Difficult Airway Management Trial (DIFFCAIR), a parallel-arm CRT that compared systematic airway assessment with existing practice for the prediction of difficult mask ventilation during anesthesia across 26 clusters (anesthesia departments) [21].

3. Results

Analyses of group-level summary data ignoring clustering led to meaningfully different SEs than the appropriate analyses and both approaches described here (Table 2).

Compared to the original, appropriate analyses, approach #1 led to similar SEs but somewhat different point estimates and 95% CIs, whereas approach #2 led to similar SEs, point estimates, and 95% CIs (Table 2).

4. Discussion

Two approaches for estimating ESSs from CRTs based on SEs from appropriate analyses taking clustering into account have been described. The first (and simplest) approach scales all counts by a single conversion factor to obtain the ESS that provides approximately the same amount of information (ie, approximately the same SE) as the appropriate analysis taking clustering into account. The second (and somewhat more elaborate) approach adjusts estimated event proportions or means and counts in each group separately to obtain group-level summary data that results in approximately the same point estimate and amount of information (SE) as the appropriate analysis taking clustering into account, thus also handling any potential statistical bias that may occur if clustering is ignored.

Both approaches were illustrated on three selected outcomes from actual CRTs using user-friendly R functions. While there were slight differences in the results obtained from the two approaches and the original appropriate

analyses, it was evident that interpretation or meta-analysis of group-level summary data after correcting ESSs using either approach is preferable to the use of the raw group-level summary data in all examples. The examples also illustrated the ability of approach #2 to (approximately) correct the summary data to obtain point estimates similar to the appropriate analysis taking clustering into account. Approach #2 can thus correct statistical bias potentially introduced from analyses not taking clustering into account, providing a potential advantage compared to approach #1 and the common design effect-approach using ICCs [1,2].

4.1. Strengths and limitations

While the outlined approaches have potential advantages to estimating ESSs using ICCs (particularly if these are not reported and must be assumed), this text and the approaches described come with certain limitations and caveats.

First, this text solely focuses on precision and statistical bias in relation to the interpretation or use of results from CRTs in meta-analyses. Considerations regarding meta-analysis of CRTs with individually randomized trials are relevant, as are issues related to risk of bias in CRTs due to, eg, knowledge of the intervention, crossovers, and spill-over effects (and bias for these reasons is likely to be bigger in magnitude than the statistical bias discussed here [22]). These considerations are beyond the scope of this text but are discussed elsewhere [1,2,22]. Second, different CRTs may use different estimands despite using the same summary effect measures [23]. While this may affect comparability and appropriateness of meta-analysis, this is not unique to this approach and similarly affects meta-analysis of CRTs using other approaches. Third, this text solely focuses on two-armed CRTs, but the approaches may be extended to CRTs with more arms. Finally, these approaches only provide approximate ESSs, and if SEs are calculated using large sample/normality approximations, this may affect the calculated ESSs differently depending on the number of clusters (particularly, with few clusters). More correct results may be obtained by meta-analyzing results from appropriate analyses taking clustering into account directly using the generic inverse variance approach when possible [2].

The described approaches also come with several strengths, including avoiding the need to assume ICCs when appropriate analysis results are reported; their reliance on well-established, simple formulae for SE calculation; the inclusion of user-friendly functions to apply both approaches using the open-source R software; and the ability of approach #2 to not only calculate ESSs to correct overly precise results but also to correct bias due to the statistical procedure used. This provides a potential advantage to approach #1 and design effect-approaches, even when ICCs are available. Finally, the two approaches can be used for both parallel-arm CRTs and extensions.

Table 2. Comparison of results using different approaches and actual versus effective sample sizes

Results	Intervention group summary data			Control group summary data			Analysis results			
PEPTIC trial: Clinically important upper gastrointestinal bleeding										
Analysis (data)	Events	Participants	Proportion	Events	Participants	Proportion	RR	95% CI	P value	SE ^a
Appropriate ^b (actual)	172	13,436	1.28%	239	13,392	1.78%	0.73	0.57 to 0.92	.009	0.1221
Summary data (actual)	172	13,436	1.28%	239	13,392	1.78%	0.72	0.59 to 0.87	<.001	0.0992
Approach #1 (effective)	114	8872	1.28%	158	8843	1.78%	0.72	0.57 to 0.91	.007	0.1220
Approach #2 (effective)	115	8862	1.30%	156	8333	1.77%	0.73	0.58 to 0.93	.012	0.1220
SuDDICU trial: days alive and free of mechanical ventilation within 90 days										
Analysis (data)	Mean	SD	Participants	Mean	SD	Participants	MD	95% CI	P value	SE
Appropriate ^c (actual)	61.9	36.1	2791	59.7	37.1	3191	2.09	−0.35 to 4.53	-	1.2449
Summary data (actual)	61.9	36.1	2791	59.7	37.1	3191	2.20	0.34 to 4.06	.020	0.9478
Approach #1 (effective)	61.9	36.1	1618	59.7	37.1	1849	2.20	−0.24 to 4.64	.077	1.2449
Approach #2 (effective)	61.8	36.1	1618	59.8	37.1	1850	2.09	−0.35 to 4.53	.093	1.2448
DIFFICAIR trial: unpredicted difficult mask ventilation										
Analysis (data)	Events	Participants	Proportion	Events	Participants	Proportion	OR	95% CI	P value	SE ^a
Appropriate ^d (actual)	427	46,804	0.91%	414	47,202	0.88%	0.98	0.66 to 1.44	.90	0.1990
Summary data (actual)	427	46,804	0.91%	414	47,202	0.88%	1.04	0.91 to 1.19	.566	0.0693
Approach #1 (effective)	52	5672	0.92%	50	5720	0.87%	1.05	0.71 to 1.55	.809	0.1990
Approach #2 (effective)	50	5651	0.88%	52	5699	0.91%	0.97	0.66 to 1.43	.876	0.1990

Comparison of results for three outcomes from three separate cluster-randomized trials according to different approaches to analysis and whether actual summary data (including actual sample sizes) or corrected summary data (including effective sample sizes) were used. The appropriate results are the reported group-level summary data (ie, actual event/participant counts) and the results from the analysis reported for each trial taking clustering into account. The summary data results are from an analysis using the reported group-level summary data using the formulae presented in Table 1, with the summary data unchanged (ie, actual event/participant counts). The approach #1 results are from the first, simple approach described in the text using the ratio between variances, including changed event/participant counts (ie, corresponding to the effective sample sizes), but unchanged proportions, means, and SDs compared to the reported summary data. The approach #2 results are from the second, approach described in the text using an optimization procedure, including changed event/participant counts (ie, corresponding to the effective sample sizes), proportions, and means, but with unchanged SDs compared to the reported summary data. The number of decimals varies due to different number of decimals used in the original reports. Comparisons with design effect approaches are not provided, as the required correlations have not been reported for the example trials, and assuming these would lead to substantial variation in results.

CI, confidence interval; DIFFICAIR, Difficult Airway Management Trial; MD, mean difference; OR, odds ratio; PEPTIC, Proton Pump Inhibitors vs Histamine-2 Receptor Blockers for Ulcer Prophylaxis Treatment in the Intensive Care Unit; RR, risk ratio (relative risk); SE, standard error; SuDDICU, Selective Decontamination of the Digestive Tract in Intensive Care Unit.

^a For relative summary effect measures, SEs are on the logarithmic scale.

^b For the PEPTIC cluster-crossover trial, results from the appropriate analysis are those reported in the primary publication, calculated using a generalized estimating equation [19]. The SE (on the logarithmic scale) was calculated from the reported 95% CI [16].

^c For the SuDDICU cluster-crossover trial, results from the appropriate analysis are those reported in the primary publication, calculated using a model with random effects [20]. The SE was calculated from the reported 95% CI [16]; a *P* value from this analysis was not reported and is hence not included in this table.

^d For the DIFFICAIR parallel-arm cluster trial, results from the appropriate analysis are those reported in the primary publication, calculated using a generalized estimating equation [21]. The SE (on the logarithmic scale) was calculated from the reported 95% CI [16].

4.2. Perspectives and future directions

These approaches are not only relevant for CRTs but can also be used to calculate ESSs from, eg, adjusted analyses of other trial designs, including individually randomized trials. This could be relevant, as there is increased attention to the precision gains when trials adjust for prognostic baseline variables [24] (ie, adjustment may increase the ESSs compared to the actual sample sizes), and meta-analysis of raw group-level summary data in such cases may reduce precision compared to adjusted analyses.

Analyses of CRTs taking clustering into account may also be adjusted for additional baseline variables. Finally, the approaches may also be used to meta-analyze results from trials using different summary effect measures by estimating the ESS using an appropriately calculated specific summary effect measure (eg, an OR) and using the corrected group-level summary data to calculate another summary effect measure (eg, an RR). The implications of using these approaches for estimating ESSs from CRTs and adjusted analyses of other trial designs compared to

alternative approaches (particularly design effect approaches) warrants further study.

5. Conclusion

Two approaches for estimating ESSs from CRTs based on reported SEs from appropriate analyses taking clustering into account have been presented, along with examples and code. Both approaches adjust the group-level summary data to ensure that simple analyses not taking clustering into account provide approximately correct SEs so that inappropriately high precision is avoided, and both approaches can be used for parallel-arm CRTs as well as extensions of CRTs. The second approach also corrects potential bias in estimates that can be introduced by ignoring clustering. These approaches may be used to aid interpretation and evidence synthesis of CRTs and can be used in cases where results from appropriate analyses of CRTs are presented, even if ICCs are not reported.

CRedit authorship contribution statement

Conceptualization: Anders Granholm. Methodology: Anders Granholm. Formal analysis: Anders Granholm. Software: Anders Granholm. Writing — original draft: AG. Writing — review and editing: Anders Granholm.

Declaration of competing interest

The author declare that they have no conflict of interest.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112200>.

Data availability

Only group-level summary data were analyzed; these are presented in the manuscript.

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